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GASTRINOMA

- Neuroendocrine tumor → Secretes gastrin.
- Neuroendocrine markers : Chromogranin, NSE.
- Clinical manifestation : Zollinger-Ellison syndrome

Types and Clinical features

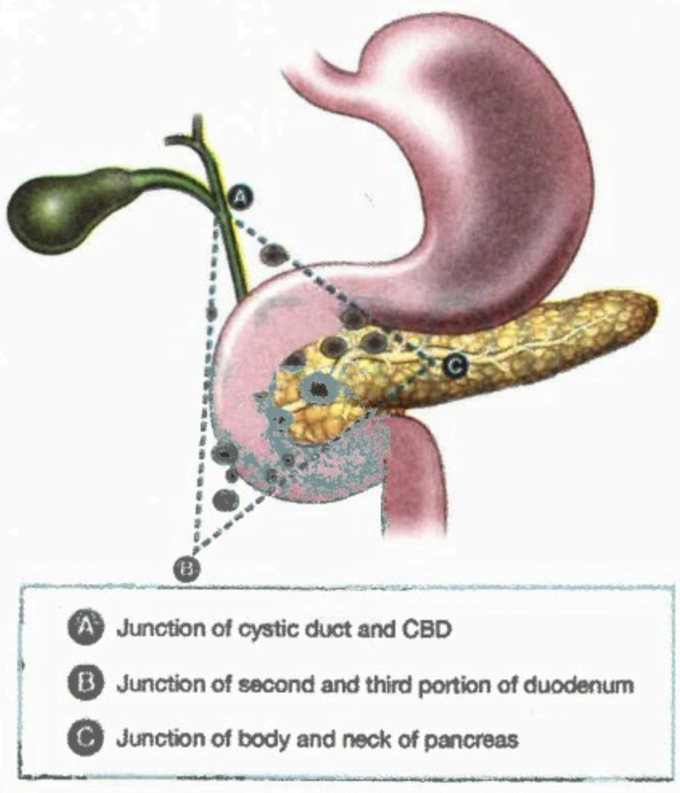
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Types :

	Sporadic (80%)	Associated with MEN-1 (20%)
Features	• 2nd m/c functional pancreatic neuroendocrine tumor. • Single, large tumor.	• m/c neuroendocrine in MEN-1. • multiple, small tumors.
Age Group	30-50 years	<20 years
Sex	F > M	M = F (Autosomal Dominant)
Malignancy risk	60% (metastasis to liver)	Low (mostly benign)
Location	Passaro's triangle/gastrinoma triangle	Duodenum
Treatment	Majority cases : Surgery	• High dose PPI. • >2.5 cm : Surgery.
Prognosis	Bad (more aggressive)	Good (Less aggressive)

Note :

- most functional pancreatic neuroendocrine tumor : Insulinoma.
- m/c cause of death in MEN-1 : Entero pancreatic neuroendocrine tumor (Gastrinoma).



Gastrinoma triangle

Effects of increased gastrin :

↑Gastrin
↓
Stimulate parietal cells
↓
↑HCl secretion
↓
Damage mucosal lining of GI tract
↓
Gastric ulcer (GU)/duodenal ulcer (DU)

Acid inactivates pancreatic enzymes :
Diarrhea, steatorrhea, malabsorption.

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Clinical features :

- most important symptom : Abdominal pain.
- most common manifestation : Peptic ulcer (Peptic Ulcer Disease).

Indicator of gastrinoma in PUD :

- Unusual location : D2 & beyond.
- Refractory to medical therapy :
  - GU >12 weeks.
  - DU >8 weeks.
- Recurrence.
- Abdominal pain : Diarrhea.

Note :

Diarrhea in gastrinoma : Both osmotic & secretory component.

Secretory : Gastrin → Acts on enterocyte → Absorb water into lumen.

Osmotic : ↑ Acid → Inactivate pancreatic enzyme → ↓ Lipolysis → Steatorrhea.

Management

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Investigation :

Fasting Gastrin levels (Screening) :

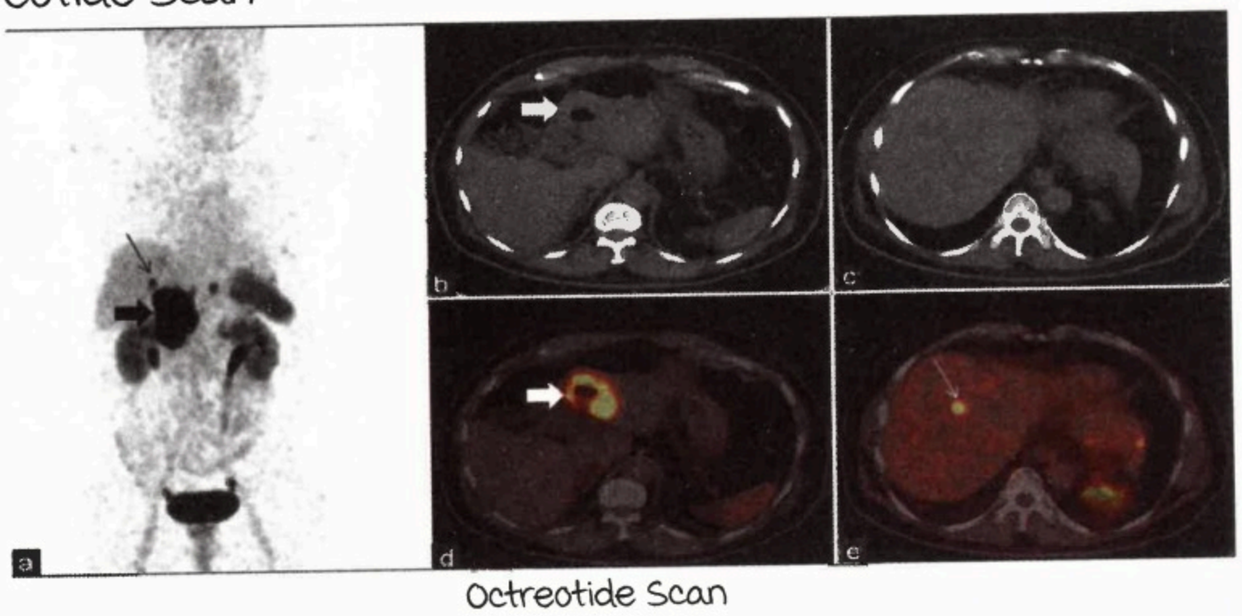
- 1000 pg/ml : Carcinoma.
- 200–1000 : Confirmatory test needed.
- <200 : Normal.

Secretin stimulation test : Confirmatory test.

- Principle : In gastrinoma, secretin stimulates gastrin (Physiologically, secretin inhibits gastrin secretion).
- Procedure : I.V secretin → measure serum gastrin at regular intervals
  ↓
  Rise by 200 pg/ml
  ↓
  Gastrinoma.

Somatostatin Receptor Scintigraphy (SRS)/Octreotide Scan :

- IOC : (most specific).
- Principle : Gastrinomas have somatostatin receptors in large numbers (Hence octreotide used in treatment also).
- Procedure : Radio active labelled octreotide given.



Octreotide Scan

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Note :

- SRS not effective in small tumor/MEN-1 → Endoscopic USG preferred.
- $\frac{\text{Basal acid output}}{\text{Maximal acid output}}$ = >0.6 : Gastrinoma.
- Investigation in pheochromocytoma : Gallium DOTATATE PET scan.

Treatment :

MEN-1 associated :

- High dose PPI (60-80mg) : Usually enough.
- If size >2.5 cm : Surgery.

Sporadic : High dose PPI + octreotide → followed by surgery.

Surgery based on location :

- Pancreatic head : Enucleation.
- Pancreatic body/tail : Distal pancreatectomy.
- Duodenum : Full thickness excision.



Surgical excision specimen

Note :

Gastrectomy : Not required.

m/c site of metastasis : Liver.